

Edge AI – A Questions and Answers Approach

**Authored by
Ning Chen, Ph.D.**

An alpha version of the textbook for Edge AI education

© 2024 Ning Chen, All Rights Reserved

(Use of any materials of this document in any educational/commercial setting requires written consent from the author)

Please note that all assignments are tentative and subject to changes.

Discussion Assignments (© 2024 Ning Chen, All Rights Reserved):

Discussion Assignment - Acknowledgment Statement

You must post (just type or copy-and-paste) the following:

I have reviewed the following:

1. CPSC 473 syllabus
2. FAQs related to discussion/essay/project assignments
3. I have reviewed <http://www.plagiarism.org/plagiarism-101/what-is-plagiarism>, and I understand the severe consequences resulting from plagiarism and missing deadlines.

(Type your name here)

Discussion Edge AI (Embedded System/AIoT/IoT/Firmware) Jobs and Their Required Skill Sets

There are many jobs in the field of Edge AI (Embedded System/AIoT/IoT/Firmware). Students who consider seeking those jobs may want to research the job wanted advertisements and analyze the required key skills and qualifications. You are required to post your findings and insights on this topic.

FAQs:

Q. Edge AI is a new buzzword. I couldn't find too many jobs with that title. What should I do?

A. The old buzzwords are embedded system development, IoT (Internet of Things), AIoT (Artificial IoT), Firmware Development. You may also want to use those keywords for your search.

Q. How do I start this discussion?

A. You should search some major job sites (e.g., Monster.com and indeed.com) with some keywords. For example, you may want to use some major keywords such as AI, Edge AI, AIoT, embedded System Development, Firmware Development ...

Q. I am not sure there are lots of AI jobs so far. May I focus on some AI startups to see what jobs they offer?

A. Yes (actually AI startups are more interesting).

Q. What should I write?

A. You can write anything as long as it relates to the job descriptions you found and their required skill sets.

Q. Should I also analyze the pay scale of those jobs?

A. Yes, if you can find enough information on the pay scale of the software testing industry.

Q. How do you grade my discussion?

A. The grading of all the discussion assignments is pass or fail (pass – 1 point, fail – 0 point).

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Q. I notice that you want us to have a section entitled “My Closing Thoughts.” May I just use the title “Closing Thoughts” without the term “My”?

A. No. This class encourages everyone to develop “independent thinking.” Your honest thinking (may not be politically correct or mainstream) is the most valuable things you can keep. Please don’t present a summary of other people’s thoughts.

Discussion What are the major differences between Edge AI and Cloud AI?

Understanding of those two terms is the starting point of our endeavor. Please research this topic.

FAQs:

Q. How do I write my research?

A. Simply search the Internet and read, read and read.

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize Chapter 2 (Edge AI in the Real World) of the textbook

FAQs:

Q. What should I post for this discussion?

A. You need to read Chapter 2 and summarize your readings.

Q. Do I need to also search the Internet to find additional information?

A. Yes (since you need to provide at least two references to meet the grading rubric).

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize Chapter 3 (The Hardware of Edge AI) of the textbook

FAQs:

Q. What should we do for this discussion?

A. You need to read Chapter 3. Please summarize your readings.

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort	At least 1000 words	Does not meet the minimum

(You can write whatever you see fit)	Required installation steps (or experiences)	
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize Chapter 4 (Algorithm for Edge AI) of the textbook

FAQs:

Q. What should we do for this discussion?

A. You need to read Chapter 4. Please summarize your readings.

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Summarize the Orange Pi 5 Plus user manual

FAQs:

Q. What should we do for this discussion?

A. You need to read the Orange Pi 5 Plus User Manual. Please summarize your readings.

Q. Where do I download the Orange Pi 5 Plus user manual?

A. https://drive.google.com/drive/folders/1Ov3mZqnMOf_8wpNt9rDxoGIR1ray2iiy

(If the above link failed, please search yourself – it is very easy to locate it.)

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion ASR (Automatic Speech Recognition)

FAQs:

Q. What should we do for this discussion?

A. You need to study/research the following:

the lecture (ppt and mp4),

whisperMLKDemo.jpynb,

LibriSpeech.jpynb

https://github.com/Uberi/speech_recognition

Q. Where do I find the above?

A. Please review the lecture (the lecture mentions all the above)

(If the github link failed, please search yourself – it is very easy to locate it.)

Q. What should I cover in my discussion?

A. See next question (basically, write something about your study/research on the subject and have “My Closing Thoughts.”

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Discussion Trying out an LLM

FAQs:

Q. What should we do for this discussion?

A. You need to study/research the following:

the lecture (ppt and mp4),

Paper: Attention Is All You Need

Paper: TinyLlama

<https://github.com/airockchip/rknn-llm>

<https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0>

Q. I need to know more about the above topics. Where do I start?

A. Please review the lecture (the lecture mentions all the above)

(If the github link failed, please search yourself – it is very easy to locate it.)

Q. What should I cover in my discussion?

A. See next question (basically, write something about your study/research on the subject and have “My Closing Thoughts.”)

Q. What is the grading rubric for the discussion assignment?

A. The grading rubric for all discussion assignments is:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words Required installation steps (or experiences)	Does not meet the minimum
Have a section labeled My Closing Thoughts	You MUST have a section labeled My Closing Thoughts	There is no section labeled My Closing Thoughts
References	At least two references	Does not meet the minimum

Group Project Task - Edge AI resources offered by TI (Texas Instruments)

FAQs:

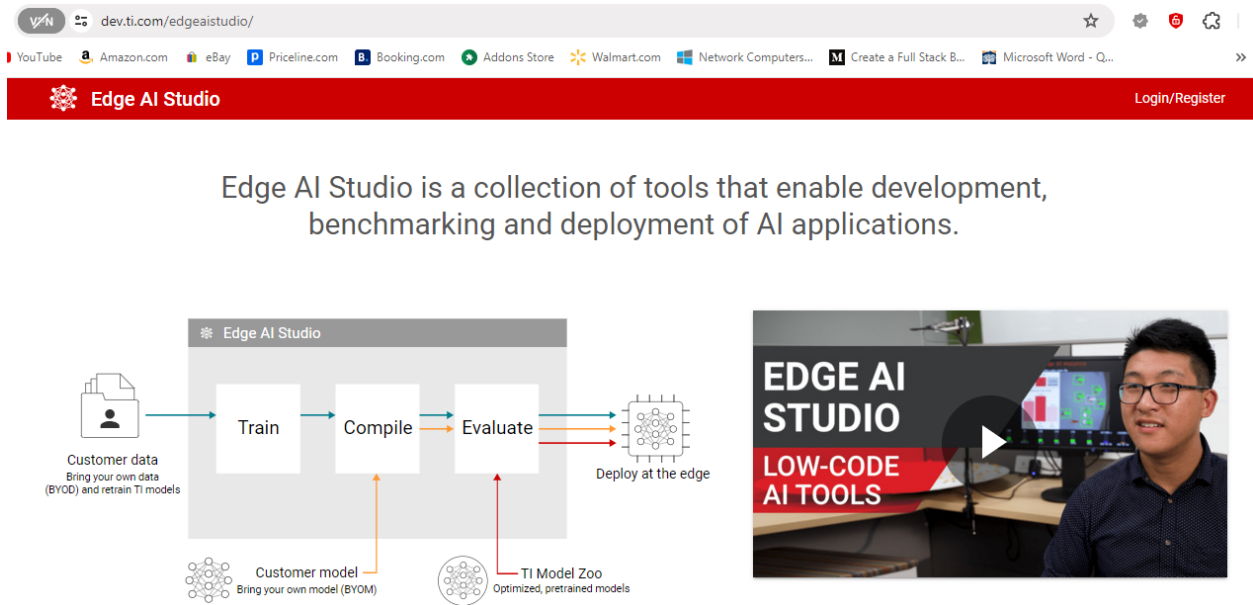
Q. What should we do for this Group Project Task?

A. Your group needs to study the Edge AI resources offered by TI.

Q. What does our group need to do?

A. Your group needs to do the following:

Step 1: Visit <https://dev.ti.com/edgeaistudio/>



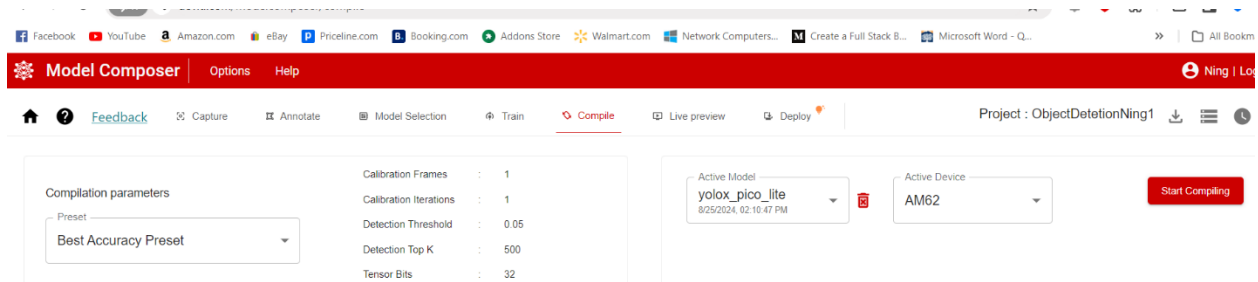
Step 2: Register a free TI account (note that your group may need to have a non-csu, business.com email account – please overcome that hurdle). Q. Our group can't register. What should we do? A. Please see me and I will assign another task for your group.

Step 3: Review two tutorials (Image Segmentation and Object Detection at <https://dev.ti.com/edgeaistudio/>)



Step 4: Play the Model Composer to reproduce two tutorials (one for Object detection and one for Image Classification).

Please note that since we don't have TI hardware board, your tutorials only needs to go through



Step 5: What to submit for this group assignment

Your group needs to submit two tutorials (two mp4 files) – one for Object Detection and the other for Image Classification. Q. What should we do in our tutorials? A. Just imitate TI's tutorials. The file names should be yourGroupNameOD.mp4 and yourGroupNameIC.mp4.

Group Project Task - Edge AI resources offered by YOLOv7 + roboflow + Google Colab

FAQs:

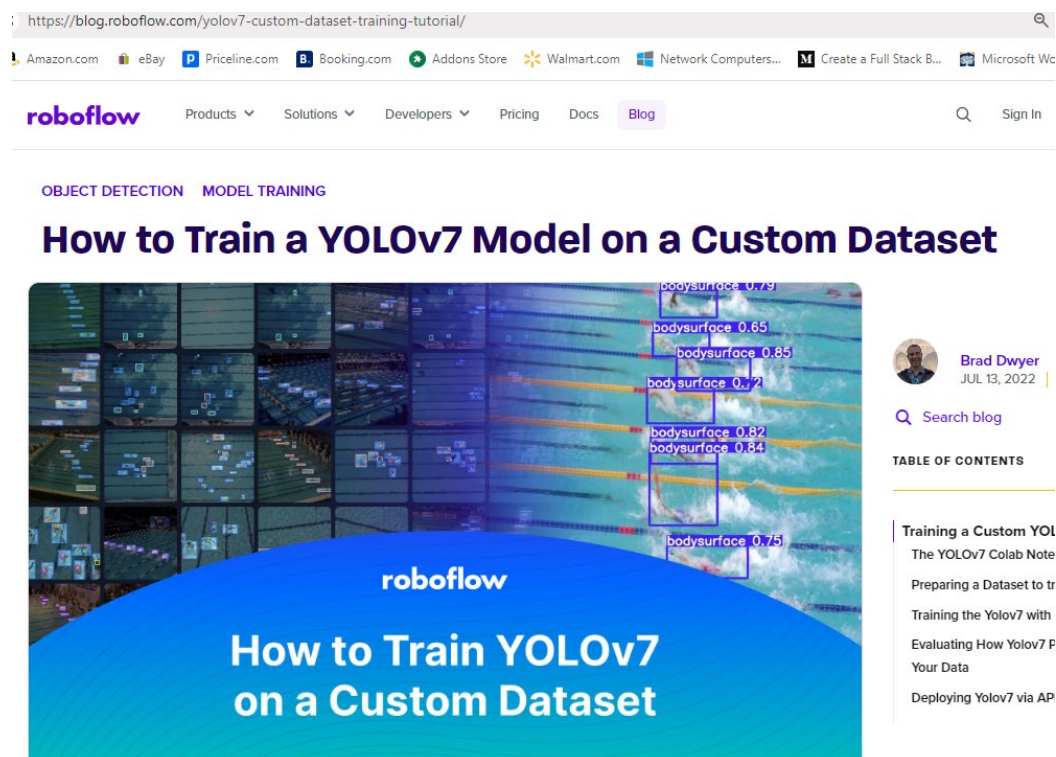
Q. What should we do for this Group Project Task?

A. Your group needs to study the Edge AI resources offered by YOLOv7 + roboflow + Google Colab.

Q. What does our group need to do?

A. Your group needs to do the following:

Step 1: Visit <https://blog.roboflow.com/yolov7-custom-dataset-training-tutorial/>



Step 2: Register a free Google account (your Google account will allow you to run google colab).

Step 3: Duplicate (or imitate) this tutorial and produce your group's own Jupyter Notebook program.

Q. Our work is almost identical to the original tutorial. Is that acceptable?

A. Yes. We try to imitate and learn from other people. Please note that your group may encounter minor issues (mostly relate to data). Please fix them yourself (not too hard).

Step 4: Submit a video (mp4) demo of your group's Jupyter Notebook. The file name should be yourGroupNmaeYOLOv7.mp4. (yes, this is the only file your group needs to submit.)

Group Project Task - Edge AI resources offered by [laitatheiYOLOv8-ONNX-RKNN-HORIZON-TensorRT-Detection](#)

FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the Edge AI resources offered by YOLOv8 model + rknn toolkit2 + python + rk3588 + orangpi 5 plus + lots of others.

Q. Wow! We are not sure we can achieve all those in a very short given time. What should we do?

A. Try to set up a playground (you need to set up rknn toolkit2) and play (if you have a playground, you can learn things naturally). Maybe we can give it a try by doing the following steps:

Step 1: Visit <https://github.com/laitathei/YOLOv8-ONNX-RKNN-HORIZON-TensorRT-Detection>

<https://github.com/airockchip/rknn-toolkit2>

https://github.com/airockchip/rknn_model_zoo

Step 2: Set up your Linux (Ubuntu) OS (under Windows OS is possible, but you are on your own)

Step 3: Duplicate and run this open-source project (by laitathe1, the first link provided above) on your own computer (laptop/desktop)

(Please review the lecture; you can install rknn toolkit2 and do the model conversion from onnx to rknn; your mp4 as an evidence should show that you have done the conversion; you can also use the onnx model to do object detection on your laptop (x86 computer) - please see the lecture, try it anyway; if failed, no worries.)

Step 4: Run the inference task on RK3588 using the rknn model (with the instructor's support in class – no need to do it yourself)

(I have done this for Groups 3 and 4, on Wednesday, 11/20/2024; I will do the same for Groups 1 and 2 after the Thanksgiving week; NO need to do this step in your group project submission)

Step 5: Each group needs to submit a presentation that includes three files (1. Report in PDF, 2. PPT file, 3. Mp4 file)

Q. What should we write in PDF?

A. PDF - must have 2000 words; some general review of the yolov8, rknn toolkit2, rk3588, My Closing Thoughts (Actually, My Group's Closing Thoughts) and at least two references.

Q. What should we write in PPT?

A. A condensed version of your PDF.

Q. What should we present in mp4?

A. About 20 minutes of video that goes through your PPT and some screen video demo (just like the class lecture).

Q. What should we include in our screen video demo?

A. You need to at least show that you can do model conversion (from onnx to rknn). This proves that your group went through the rknn toolchain installation and usage. I encourage you to do

object detection of a given photo on your laptop directly in the toolchain environment (my lecture mp4 has demonstrated it; you may need to download rnkk model zoo). Please try it (if it failed, that's fine - no deduction of grades).

Group Project Task - Handwritten Digit Recognition (MNIST classification with CNN on Colab)

FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study MNIST handwritten digits with CNN on Colab.

Q. Wow! That's exciting but very demanding (we know nothing about MNIST + CNN + Pytorch + Colab). Where do we start?

A. Try to set up a playground and play (if you have a playground, you can learn things naturally). Maybe we can give it a try by doing the following steps:

Step 1: Visit and Read

URL 1:

https://www.youtube.com/watch?v=gBw0u_5u0qU

PyTorch Course (2022), Part 4: Image Classification (MNIST)

URL 2

https://github.com/lukepolson/youtube_channel/blob/main/Python%20Tutorial%20Series/pytorch4.ipynb

URL 3

<https://github.com/github-bowen/BUAA-ML-MNIST-Classification/blob/main/MNIST-classification-with-CNN/train-on-google-colab.ipynb>

Q. Can we find other URLs and learn more?

A. Yes. Of course.

Step 2: Imitate

In Step 2 you need to run the train-on-google-colab.ipynb (jupyter notebook) on your own google colab account.

Q. Is hard to learn colab?

A. No (but no way to avoid dumping hours – it is a good investment).

Q. Will train-on-google-colab.ipynb run on my google colab account with no issues?

A. No. You do need to resolve some issues yourself.

Q. Do you have a read-to-run train-on-google-colab.ipynb? If yes, can you give it to us?

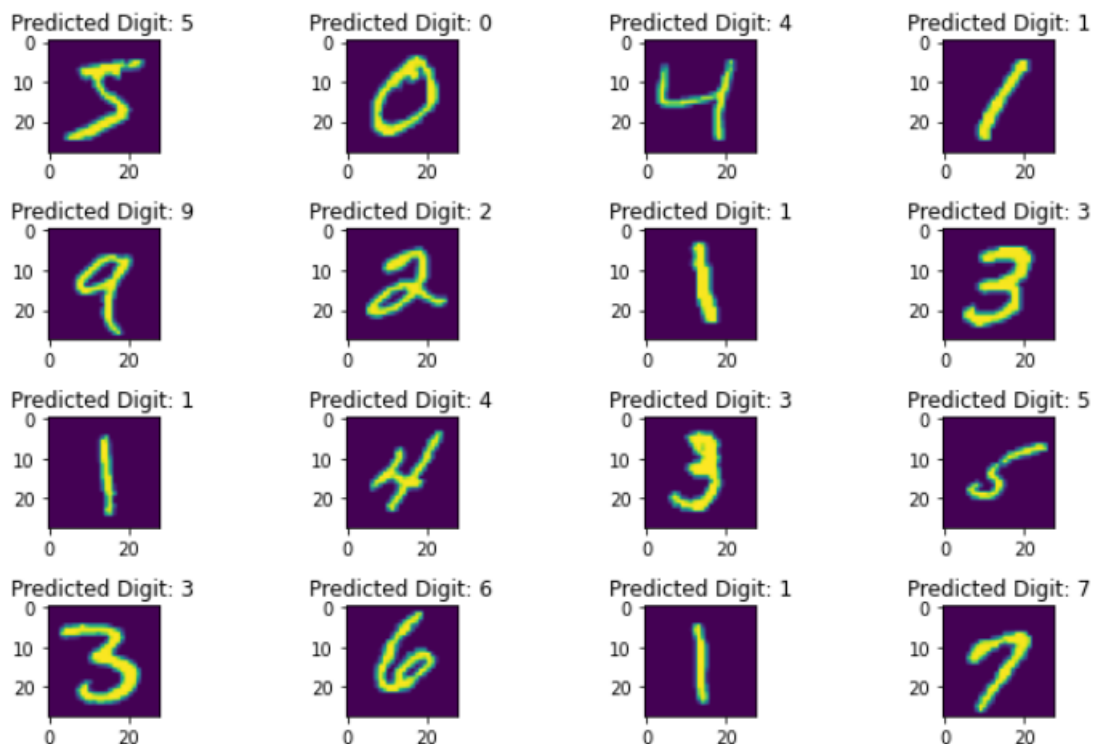
A. Yes. I do have one (after about 10 hours of fixing). Sorry, I don't think giving it to you helps you in any way. Struggling through it is probably the most effective/fastest way of learning.

Step 3: Innovate (just one baby step)

In Step 3 you need to improve the train-on-google-colab.ipynb (you can change its name as you see fit/like) a bit on your own google colab account.

Q. What kind of improvements?

A. URL 2 gives some nice plots (for example, see below). You need to add some of those plots to your jupyter notebook (as you see fit/like – it is your task; you are the boss).



Step 4: Innovate (another small baby step)

In Step 4, you need to improve your version of the jupyter notebook a small step further.

Q. What is another small baby step?

A. You need to save the pytorch model (you trained before) in your local drive (your laptop/computer) first. Add a new section in your notebook to read the saved model (pre-trained model). Run several test cases using the pre-trained model.

Q. What is the purpose of Step 4?

A. After finishing Step 4, you can use other people's pre-trained models to make interesting inferences.

Step 5: Deliverables

In Step 5, you (your group) need to submit the following:

Your jupyter notebook, mp4 (video) that demonstrates the running of your jupyter notebook.

Group Project Task - Training a Classifier (CIFAR10 dataset with CNN on Colab)

FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the tutorial of [CIFAR10 dataset with CNN on Colab](#).

Q. Wow! That's exciting but very demanding (we know nothing about CIFAR10 + CNN + Pytorch + Colab). Where do we start?

A. Try to set up a playground and play (if you have a playground, you can learn things naturally). Maybe we can give it a try by doing the following steps:

Step 1: Visit and Read

URL 1: https://pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html

(You are welcome to search/read other URLs you can find.)

Step 2: Download the Jupyter Notebook file of this tutorial to your local computer. You can use the download link:

https://pytorch.org/tutorials/_downloads/4e865243430a47a00d551ca0579a6f6c/cifar10_tutorial.ipynb (If the link fails, please use URL 1 – you can find the download link on that page.)

Q. Can I use my Google Drive to save the file (instead of using my own local computer)?

A. It is not recommended. For this assignment, please save the Jupyter Notebook file on your own local computer (since you can study/modify it easily).

Step 3: Run the Jupyter Notebook file on your colab account

Q. How do I do that?

A. There are many different ways. An easier way probably is to log in to your gmail account first and then use your browser to search for colab (and click the link, then you will be on colab).

Q. How do I upload my local Jupyter Notebook file to colab?

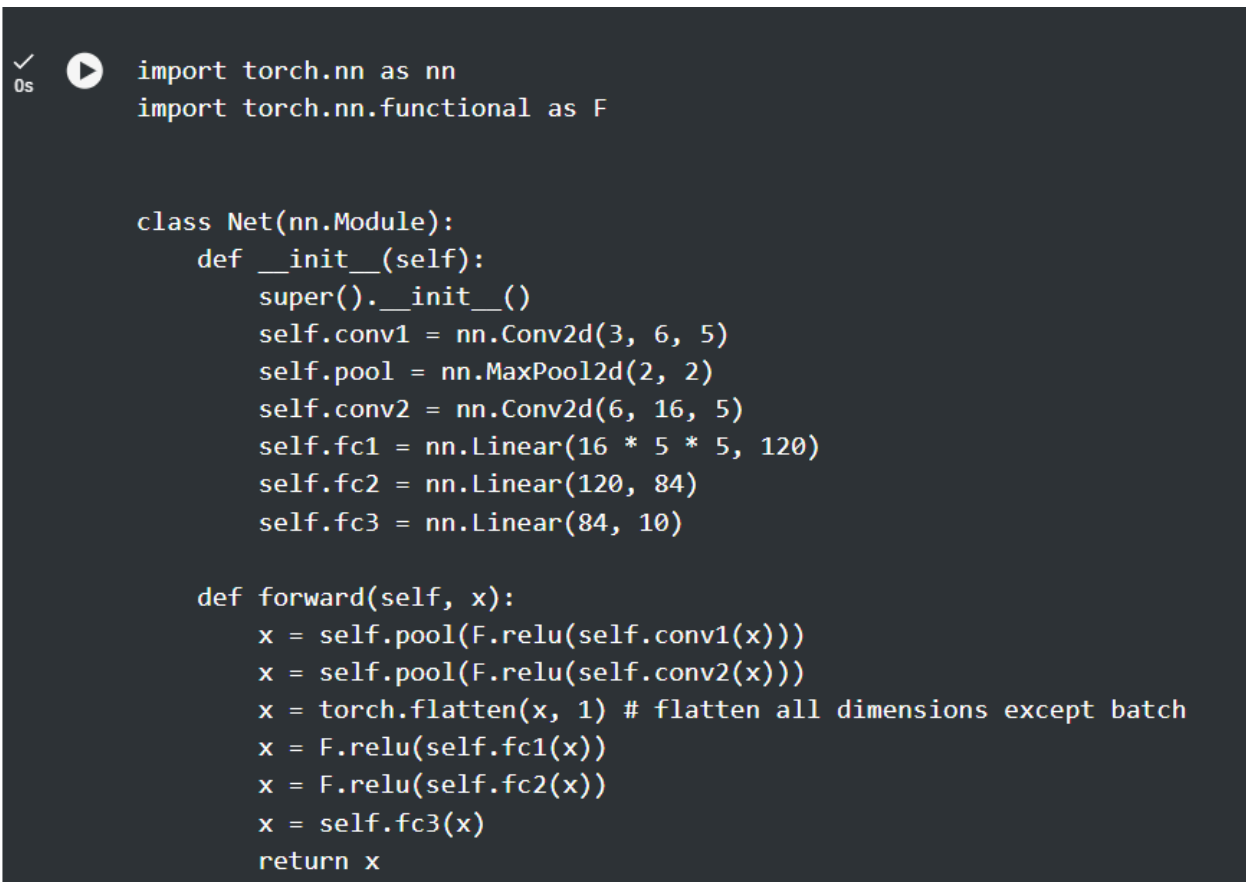
A. File -> Upload notebook

Q. How do I run my notebook file?

A. Please just google a bit.

Step 4: Run the notebook (it is bugfree; very easy to run – just enjoy it)

Step 5: Study/research the class net (shown below)



```
✓ 0s  import torch.nn as nn
import torch.nn.functional as F

class Net(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 6, 5)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(6, 16, 5)
        self.fc1 = nn.Linear(16 * 5 * 5, 120)
        self.fc2 = nn.Linear(120, 84)
        self.fc3 = nn.Linear(84, 10)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = torch.flatten(x, 1) # flatten all dimensions except batch
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = self.fc3(x)
        return x
```

Step 4: Deliverable

Each group needs to submit a research paper (in pdf) that teaches readers about this class.

Q. What should our group write?

A. There are many things your readers may not understand. For example, they may not have any ideas on Conv2d, MaxPool2d, Linear (fc1, fc2, fc3, why there are three of them), pool, relu, flatten.

Q. Can we include figures and photos?

A. Yes.

Q. Any minimum requirements?

A. Yes. Your report should have at least 5 pages long – title page won't count (more pages are welcome).

Q. Any required format/style?

A. No. You can write whatever you see fit.

Q. Do we need to say anything about the rest (running the notebook, image classification, training, accuracy, etc)?

A. No. In this assignment, we only focus on the model architecture (structure) itself.

Group Project Task - Drone Connection (DJI Tello Experiments)



FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the tutorial of <https://github.com/hanyazou/TelloPy> as the first step.

Q. What is the second step?

A. You need to write a simple doneConnect.py in python that connects to the drone (DJI Tello).

Q. How do I (your group as a whole) write this program?

A. Easy! It is given below:

```
from djitellopy import tello
import time

me = tello.Tello()
me.connect()
print("Ning Test: Battery Status")
print(me.get_battery())
```

Q. I don't have a DJI Tello drone. How do I test my doneConnect.py?

A. The instructor will bring a DJI Tello to the classroom. Each group can then test its droneConnect.py.

Q. How to test (I have no idea)?

A. If you need advice, the class instructor recommends (not required) the following items:

Item 1: Turn on Tello (the DJI drone) – Tello will be seen as a wifi SP (service point). (The instructor will turn on Tello for you.)

Item 2: Turn on you laptop's wifi to search Tello. Once found, connects to it.

Item 3: Run your droneConnect.py under VScode in the debug mode (stepping through all the source code and make sure you don't have any syntax/package/module issues).

Item 4: Make sure you turn all OS firewall off (in most case, making Tello SP as a private network may solve all firewall-related issues)

Item 5: If dronConnect.py runs, you will see the battery status of the Tello drone.

Q. Do I need to turn in anything in this group assignment?

A. Yes. Each group (one copy per group with group member names) needs to submit a written report in PDF that contains the following:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words of anything you like to write	Does not meet the minimum

Have a section labeled Our Closing Thoughts	You MUST have a section labeled Our Closing Thoughts	There is no section labeled Our Closing Thoughts
References	At least two references	Does not meet the minimum

Group Project Task - Drone AI - Face Detection (DJI Tello Experiments)



FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the following URLs as the first step:

<https://github.com/cvzone/cvzone>

Q. What is the second step?

A. You need to test the following drone control program:

```
from djitellopy import tello
import time

me = tello.Tello()
me.connect()
print(me.get_battery())
```

```
me.takeoff()

# Move using Distance
me.move_up(80)

# Move using Speed
me.send_rc_control(0, 0, 0, 20)
time.sleep(5)
me.send_rc_control(0, 0, 0, 0)

me.land()
```

Q. I don't have Tello. How do I test the drone control program?

A. You need to test your drone control program in the classroom. The instructor will prepare Tello as an SP (wifi service point). Your drone control program then can control Tello.

Q. That's it?

A. No. We have two more steps to go. The third step is to study the following face detection program:

```
from djitellopy import tello
import cv2
from cvzone.FaceDetectionModule import FaceDetector

me = tello.Tello()
me.connect()
print(me.get_battery())
me.streamoff()
me.streamon()

detector = FaceDetector()

while True:
```

```

img = me.get_frame_read().frame
img, bboxes = detector.findFaces(img, draw=True)
cv2.imshow("Image", img)
if cv2.waitKey(5) & 0xFF == ord('q'):
    break
cv2.destroyAllWindows()

```

Q. Can I also test the face detection program?

A. Yes. You can do the test in the classroom.

Q. What is the task I need to achieve in Step 4 (the last step)?

A. Step 4 is a bit challenging. You need to study cvzone (see URL mentioned before) first. Then, you need to develop a program (called faceDetectJump.py). In this program, the drone takes off first and detects a face. Once detected, the drone flies a bit higher for 3 seconds and then back down to the original elevation. When the key q (on the laptop) is pressed, the drone lands.

Q. Should I test my faceDetectJump.py?

A. Yes. You do have a chance to run your faceDetectJump.py in the classroom on Tello.

Q. Do I need to turn in anything in this group assignment?

A. Yes. Each group (one copy per group with group member names) needs to submit a written report in PDF that contains the following:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words of anything you like to write	Does not meet the minimum
Have a section labeled Our Closing Thoughts	You MUST have a section labeled Our Closing Thoughts	There is no section labeled Our Closing Thoughts
References	At least two references	Does not meet the minimum

Group Project Task - Robot AI - Connection (Robot Experiments)



FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the following URLs as the first step:

<https://www.hiwonder.com/collections/ros-robot/products/ainex?variant=40257678147671>

<https://www.hiwonder.com.cn/store/learn/123.html>

<https://ros.org/>

Q. Too many PDFs to read. We have tried. What next?

A. Thanks (no way to ready everything). Try to practice just-in-time reading. After turn-on, the robot enters AP (Access Point with the name HW-DDD****) mode. This means it becomes a wifi service/access point. Your laptop can discover it and connect to it.

Q. Ok, my OS wifi connects to the robot. What next?

A. You need to run an SSH “terminal” program on your laptop. A free terminal program MobaXterm is recommended (well, since HiWonder, the robot’s provider recommended it). You can easily

download MobaXterm (portable version is the best to use) from <https://mobaxterm.mobatek.net/download.html> (or search the link yourself). In Windows 11, you need to click – Start -> All Apps -> MobXterm

Q. Okay, now I can ssh to the robot. What kind of things I can do?

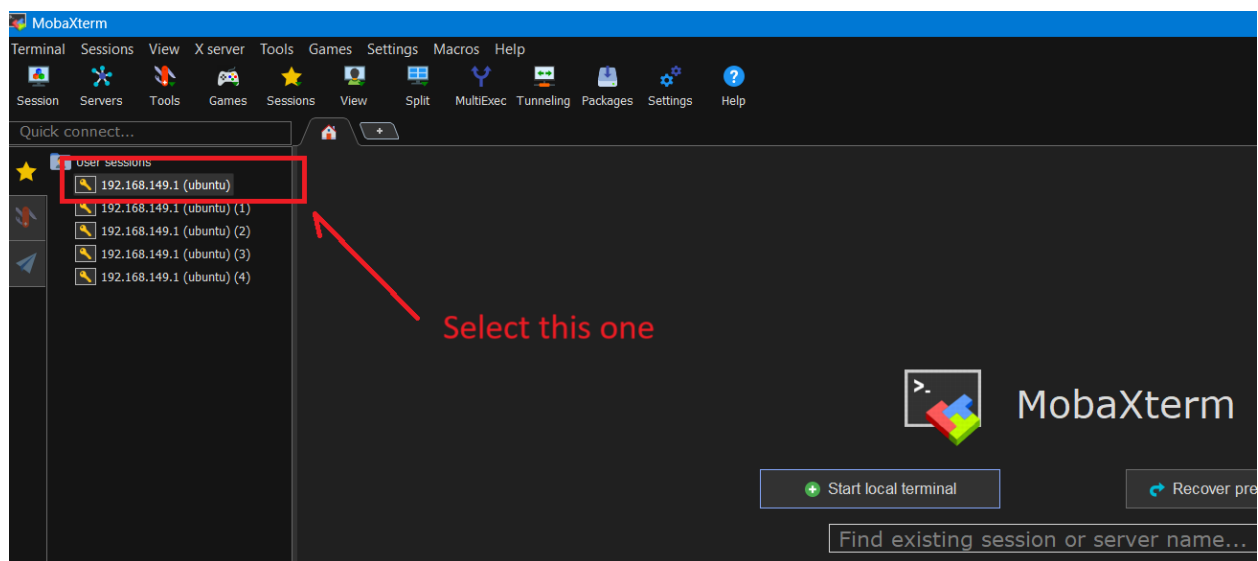
A. The robot OS is ubuntu. On your laptop, you can ssh to the robot OS (via MobaXterm). You can cd to /home/ubuntu/software/collect_picture folder.

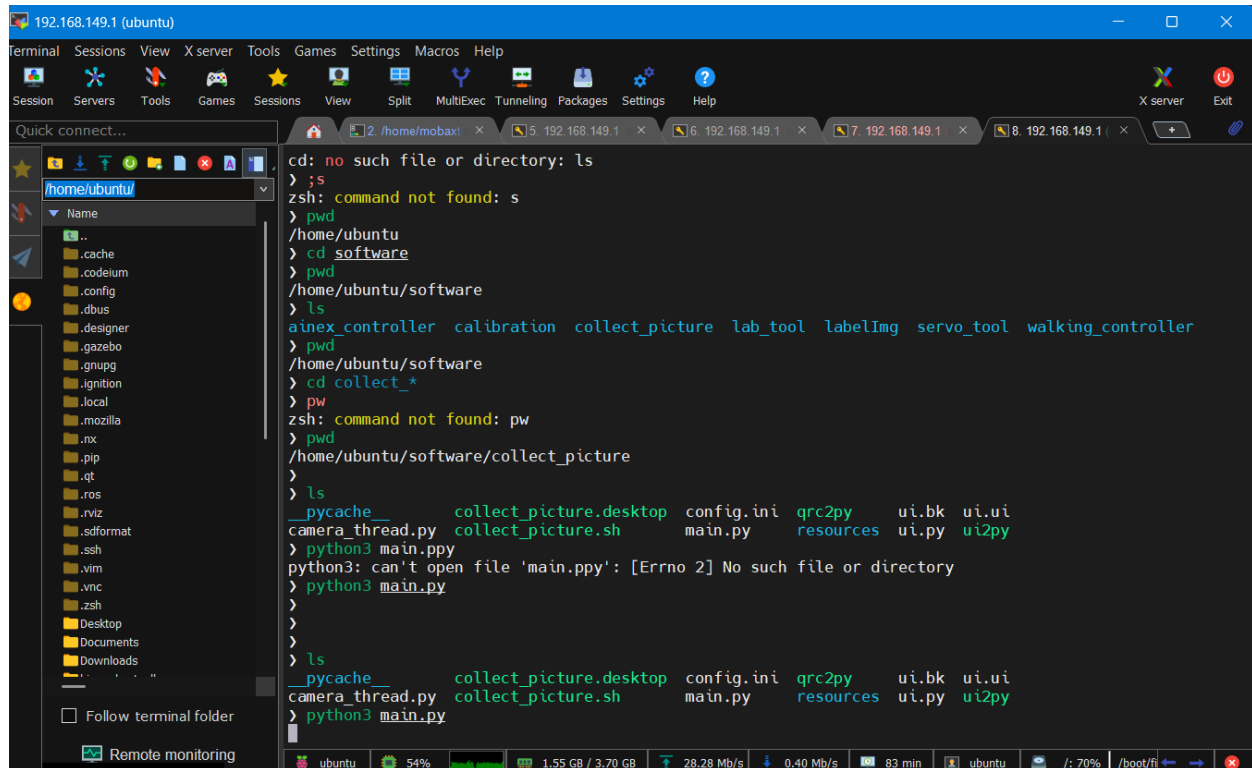
Then type: python3 main.py (run the main.py program in that folder).

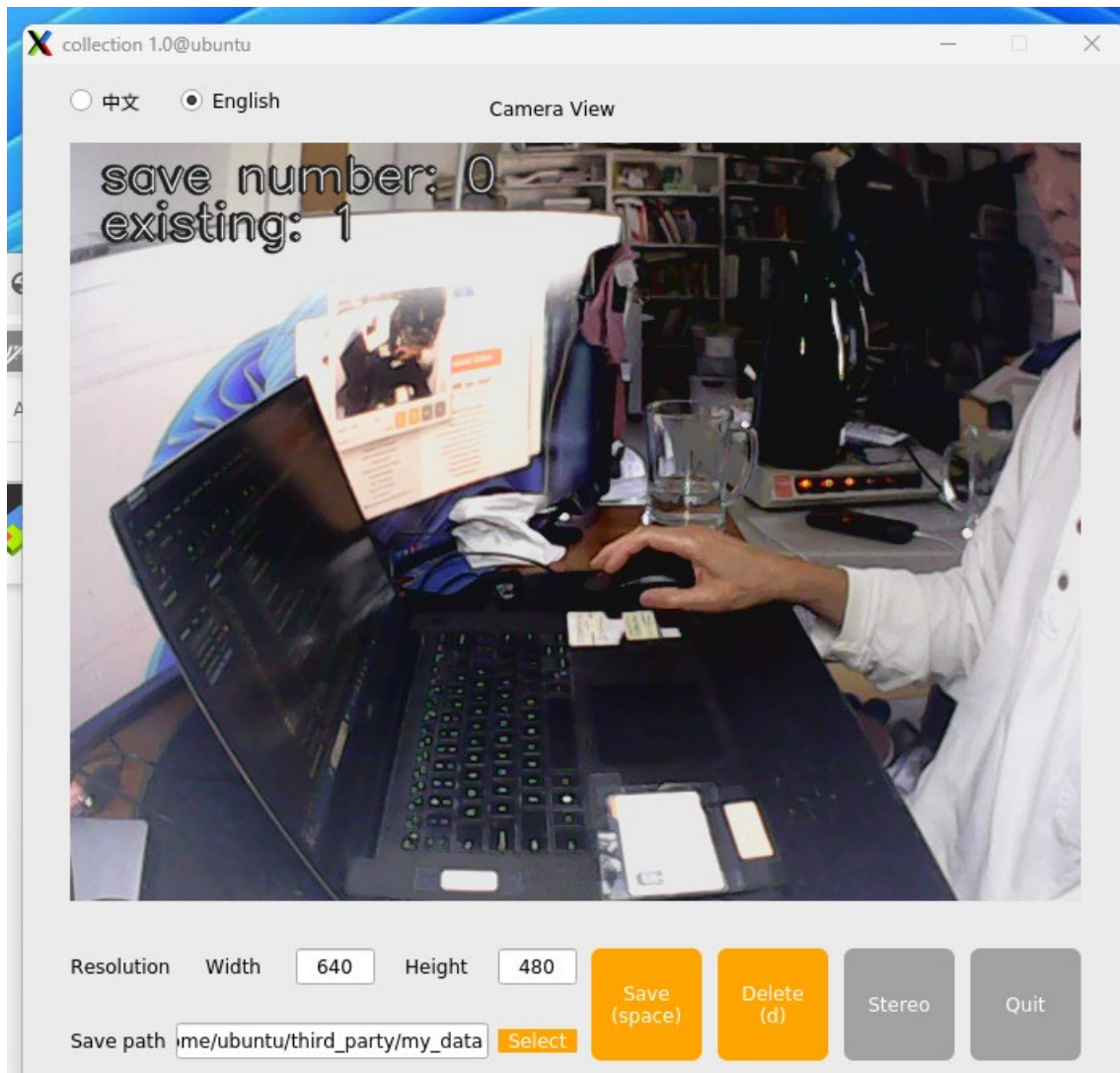
You will see the picture (streaming video) from the robot's camera on your laptop.

Q. I don't have the robot. How do I execute the above steps?

A. Your group needs to install MobaXterm on your laptop. The instructor will bring the robot to the classroom and we can try it there.







(The robot's camera took the above photo.) The whole folder of collect_picture is available from the instructor. If you want to learn how to stream video from one Ubuntu (in our case, Raspberry Pi 4) to your laptop, I think this python program is a great start.

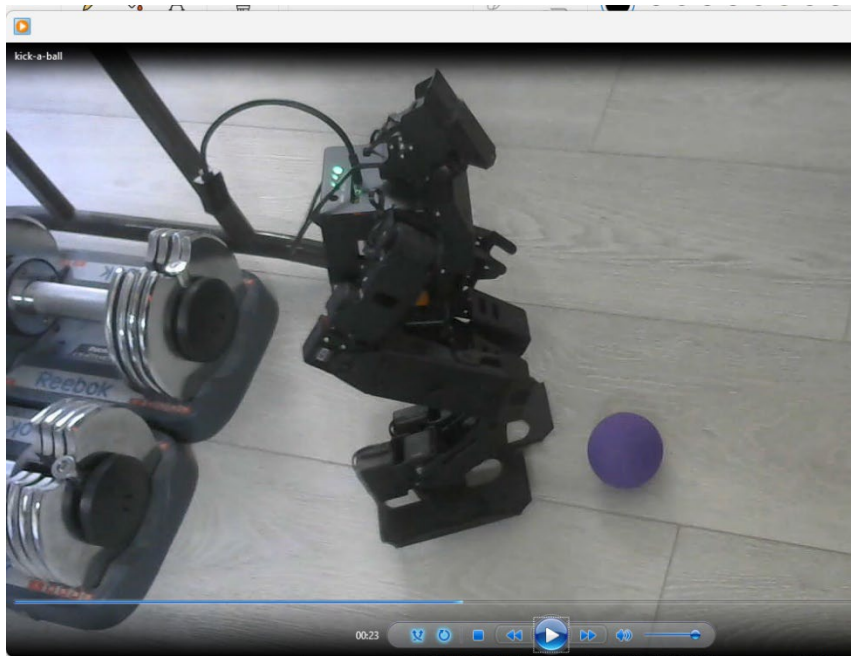
Q. Do I need to turn in anything in this group assignment?

A. Yes. Each group (one copy per group with group member names) needs to submit a written report in PDF that contains the following:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words of anything you like to write	Does not meet the minimum

Have a section labeled Our Closing Thoughts	You MUST have a section labeled Our Closing Thoughts	There is no section labeled Our Closing Thoughts
References	At least two references	Does not meet the minimum

Group Project Task - Robot AI - Kick a Ball (Robot Experiments)



Q. What should we do for this Group Project Task?

A. Your group needs to study the following resources as the first step:

Video (kick a ball, a video on the class website)

Python program (on the class website)

Q. What next?

A. The instructor will bring the robot to the classroom. Each group can run the kick-a-ball python program via your laptop's MobaXterm.

Q. How?

A. Do the following via MobaXterm:

```
cd /home/ubuntu/ros_ws/src/ainex_example/scripts/kick_ball
```

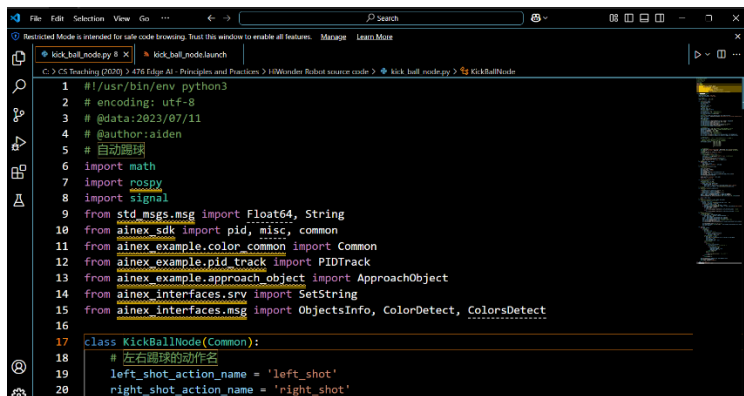
```
sudo systemctl stop start_app_node.service
```

```
roslaunch ainex_example kick_ball_node.launch
```

You will see the robot in action.

Q. We are just rerun some existing python program to achieve the task of kick-a-ball. Can we do anything to learn a bit more?

A. Yes. The first step is to imitate some working example (like, kick-a-ball). In this step, we need to study the source code of kick_ball_node.py (on the class website).



```
1 #!/usr/bin/env python3
2 # encoding: utf-8
3 # @data: 2023/07/11
4 # @author: aiden
5 # 自动踢球
6 import math
7 import rospy
8 import signal
9 from std_msgs.msg import Float64, String
10 from ainex_sdk import pid, misc, common
11 from ainex_example.color.common import Common
12 from ainex_example.pid_track import PIDTrack
13 from ainex_example.approach_object import ApproachObject
14 from ainex_interfaces.srv import SetString
15 from ainex_interfaces.msg import ObjectsInfo, ColorDetect, ColorsDetect
16
17 class KickBallNode(Common):
18     # 左右踢球的动作名
19     left_shot_action_name = 'left_shot'
20     right_shot_action_name = 'right_shot'
```

The screenshot shows a MobaXterm terminal window with a file explorer on the left and a terminal pane on the right. The file explorer shows the directory `/home/ubuntu/software/collect_pi` with files like `__pycache__`, `resources`, `camera_thread.py`, `collect_picture.desktop`, `collect_picture.sh`, `config.ini`, `main.py`, `qrc2py`, `ui.bk`, `ui.py`, `ui.ui`, and `ui2py`. The terminal pane shows the following code and commands:

```
2.LINE_AA

    frame_rgb = image_resize
    qimage = QImage(frame_rgb.data, frame_rgb.shape[1], frame_rgb.shape[0], QImage.Format_RGB888)
    qpix = QPixmap.fromImage(qimage)
    self.label_display.setPixmap(qpix)
except BaseException as e:
    print(e)

if __name__ == "__main__":
    app = QApplication(sys.argv)
    rospy.init_node('collect_picture_node')
    myshow = MainWindow()
    myshow.show()
    sys.exit(app.exec_())
python3 main.py

> cd /home/ubuntu/ros_ws/src/ainex_example/scripts/kick_ball
> ls
kick_ball_node.launch  kick_ball_node.py
> sudo systemctl stop start_app_node.service
> roslaunch ainex_example kick_ball_node.launch
> sudo systemctl stop start_app_node.service
> roslaunch ainex_example kick_ball_node.launch
... logging to /home/ubuntu/.ros/log/8a1c7a00-64bc-11ee-86ab-496e4fa6ab18/roslaunch-ubuntu-49923.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://ubuntu:35139/

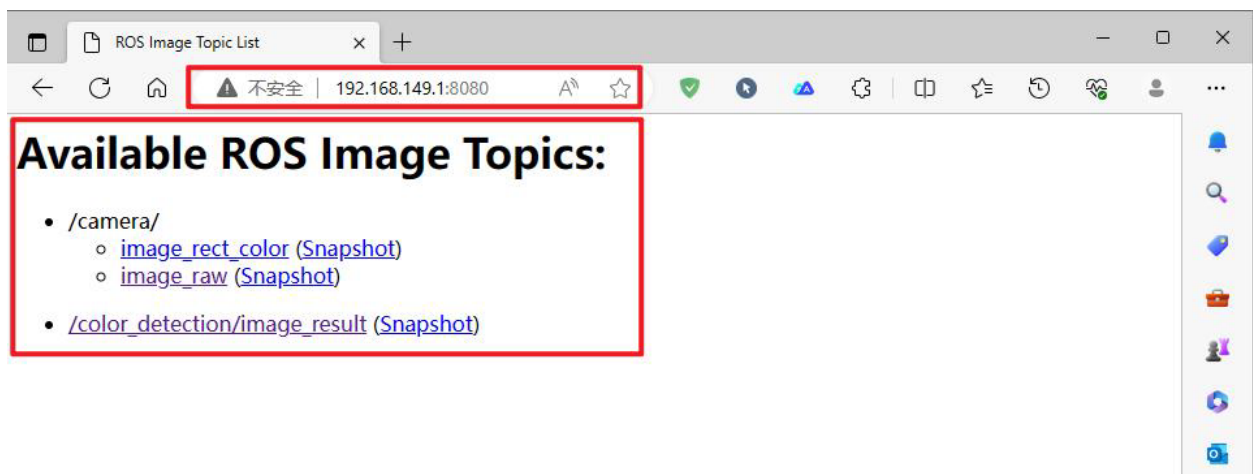
SUMMARY
=====
```

Q. Can we see the image seen by the robot's camera on our laptop?

A. Yes.

After launching the game, as the MobaXterm terminal cannot display the live camera feed, you can address this by enabling the 'web_video_server' service to view the camera feed in a web browser.

1) Open any web browser, and enter the address, 192.168.149.1:8080



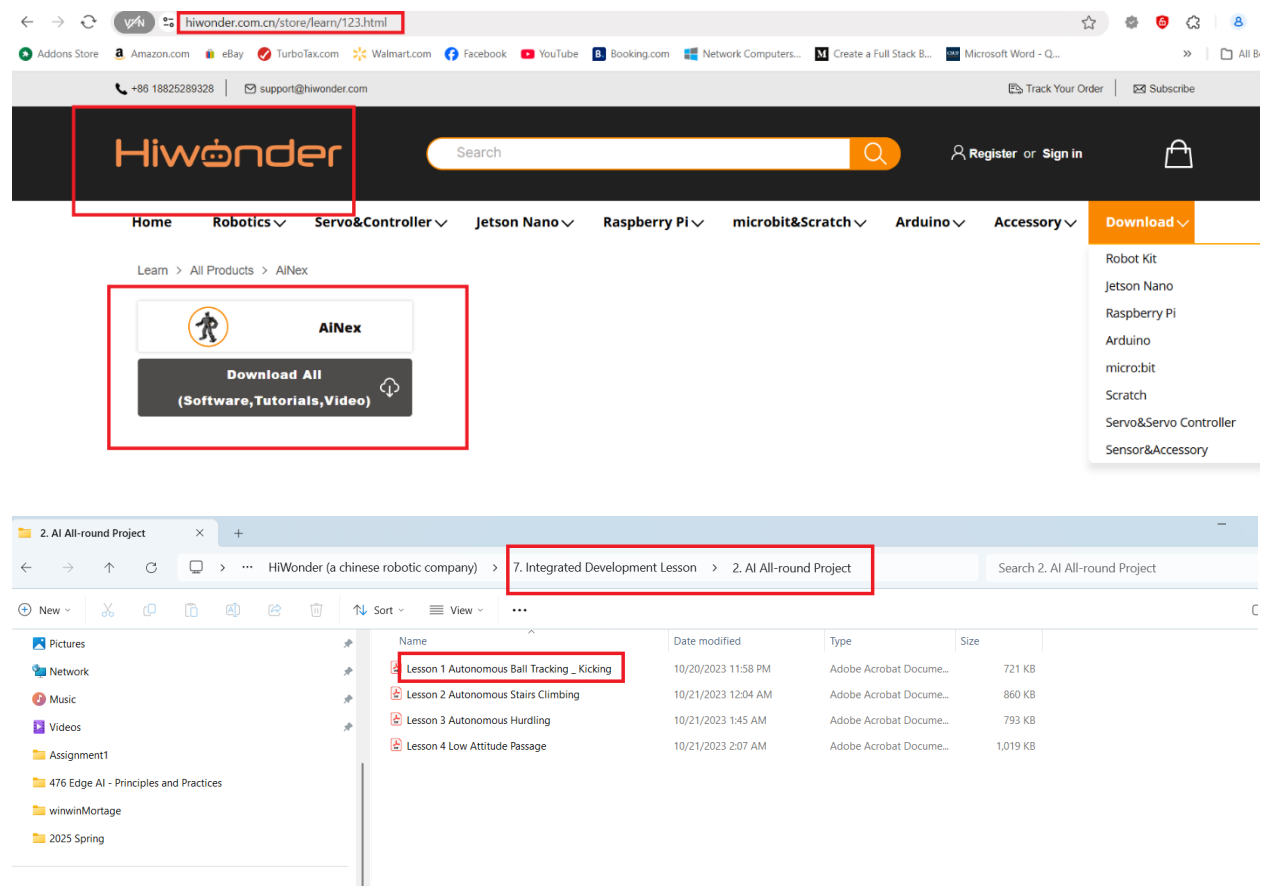
2) Select “/color_detection/image_result” to access the live camera feed.

Q. How to terminate the kick_ball program?

A. Control-C will terminate the program.

Q. I want to read more. Where can I find more information?

A. You can visit hiwonder.com and search for AiNex resources (Tutorials and Software).

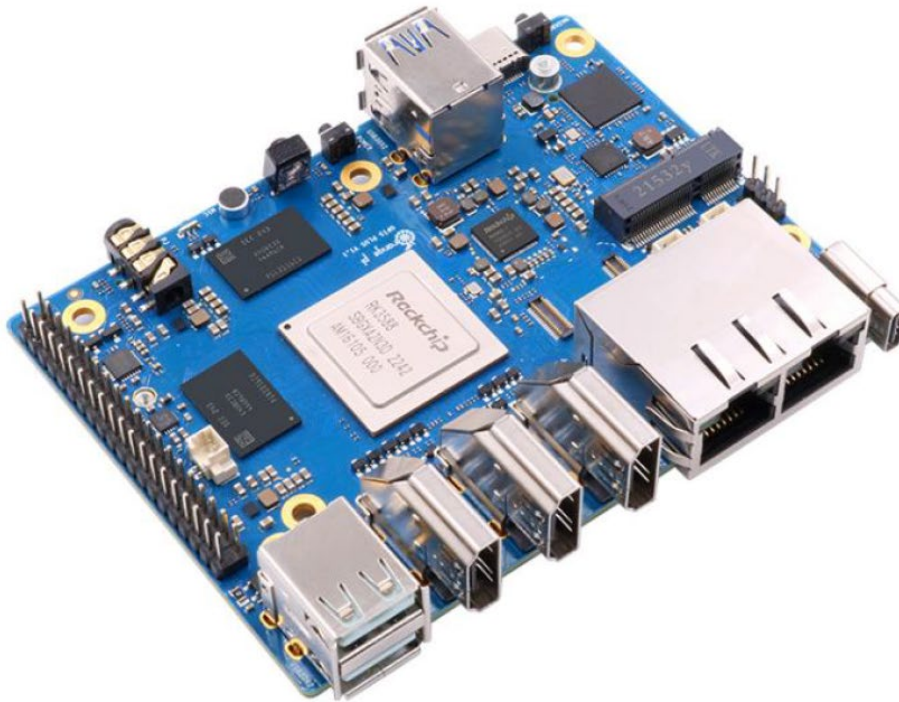


Q. Do I need to turn in anything in this group assignment?

A. Yes. Each group (one copy per group with group member names) needs to submit a written report in PDF that contains the following:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words of anything you like to write	Does not meet the minimum
Have a section labeled Our Closing Thoughts	You MUST have a section labeled Our Closing Thoughts	There is no section labeled Our Closing Thoughts
References	At least two references	Does not meet the minimum

Group Project Task - Loading OS to the Orange Pi 5 Plus Board



FAQs:

Q. What should we do for this Group Project Task?

A. Your group needs to study the Orange Pi 5 Plus User Manual (provided on the class Canvas site) as the first step.

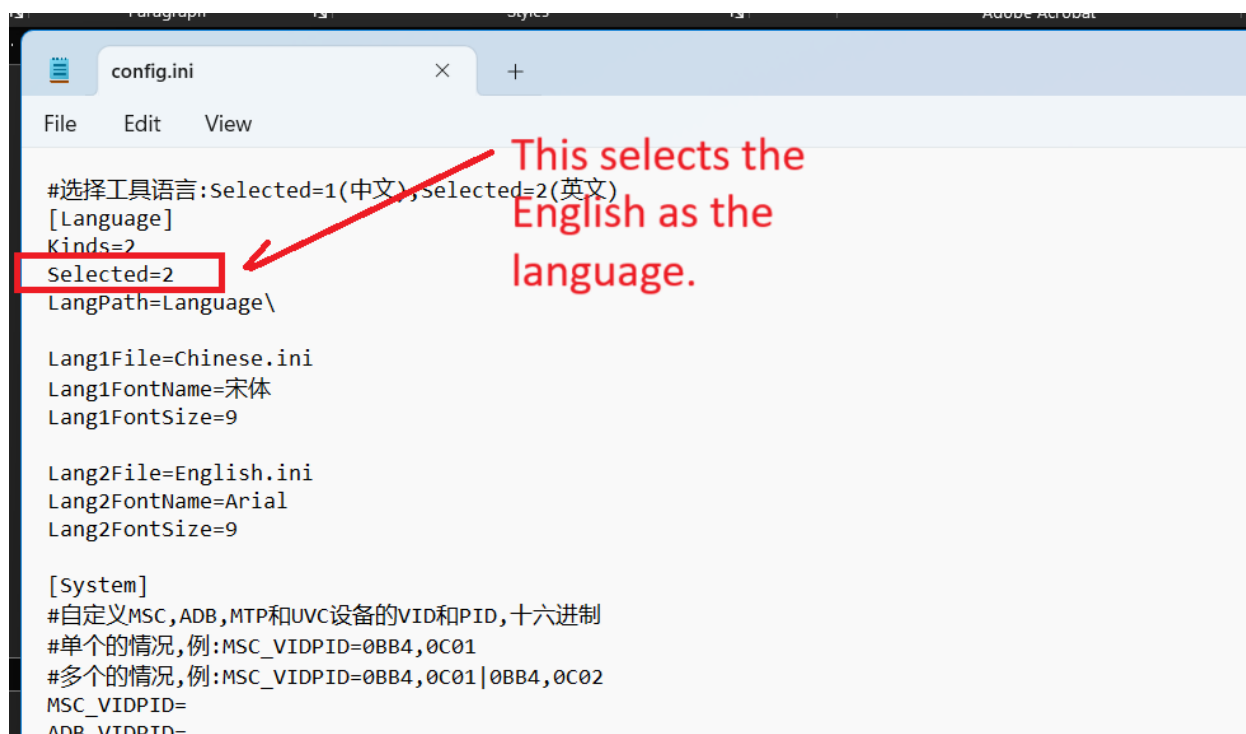
Q. What is the second step?

A. You need to study the Section 2.5 (How to burn Linux image to eMMC) of the Orange Pi 5 Plus User Manual and roughly know the workflow for installing Unbutu OS.

Q. What is the third step?

A. download Rockchip DriverAssitant_v5.12.zip and MiniLoader and the burning tool RKDevTool_Release_v3.15.zip from the

Orange Pi data download page (Or, get it from the class Canvas site). Please read Section 2.5 carefully. Please also note that in the config.ini file of the RKDevTool_Release_v3.15 folder, you want to change the language setting (to English):



Q. What is the fourth step?

A. You need to download Ubuntu 24.04 LTS image (for the ARM processor, NOT for the x86 Intel processor) from Github (<https://github.com/Joshua-Riek/ubuntu-rockchip/wiki/>).

Q. What is the fifth step?

A. The fifth step will be done in the classroom with all the hardware provided by the instructor. Since each group may only have about 20 minutes, your group may want to perform the steps 1 to 4 well in advance before the class time. One more time, loading OS to the board does require careful attentions (read the manual well!).

Q. What is the end result of this group task?

A. If things go well, you can see that Ubuntu 24.04 runs on your board (you need to connect the monitor, keyboard, mouse to the board first).

Q. Do I need to turn in anything in this group assignment?

A. Yes. Each group (one copy per group with group member names) needs to submit a written report in PDF that contains the following:

Category	Pass (1 point)	Fail (0 point)
Minimum Effort (You can write whatever you see fit)	At least 1000 words of anything you like to write	Does not meet the minimum

Have a section labeled Our Closing Thoughts	You MUST have a section labeled Our Closing Thoughts	There is no section labeled Our Closing Thoughts
References	At least two references	Does not meet the minimum